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🌐 romia.dev · Portfolio with case studies, architecture breakdowns, and interactive demos

Summary

AI Engineer building production multi-agent and RAG systems, with full-stack breadth behind them: FastAPI and Spring Boot backends, event-driven microservices with Kafka, and React/Next.js frontends. Co-founded Zaylon AI and built its AI system and backend: a LangGraph tool-calling agent and tri-lingual NLP on a multi-tenant FastAPI backend.

Experience

Software Engineer & Co-Founder

May 2025 – Present

Zaylon AI

- Led the engineering team at Zaylon AI; own architecture, code reviews, technical roadmap, and DevOps.
- Built the **FastAPI** Python backend and service architecture with **Docker** and **GitHub Actions** CI/CD.
- Built the conversational AI in **LangGraph** as a single tool-calling agent with ~30 scoped tools (consolidated from an earlier multi-agent supervisor for reliability).
- Built hybrid product search: keyword + vector retrieval with reciprocal rank fusion over the catalog.
- Engineered **DialectBridge**: tri-lingual NLP with auto dialect detection for English, Egyptian Arabic, and Franco-Arabic.
- Hardened the backend: **argon2id** API-key auth, row-level multi-tenant isolation across ~80 tables, and a Postgres saga engine with compensation for checkout.

Software Integration Engineering Intern

Jul 2025

Ejada

- Built **Spring Boot** RESTful APIs for core banking integration with query optimization and **Redis** caching.
- Configured **Apache Kafka** producers and consumers for async event-driven messaging and service decoupling.
- Implemented API gateway security (OAuth2, rate limiting, request validation) using **WSO2 API Manager**.

AI Coding Expert

Sep 2024 – Feb 2026

Freelance

- Reviewed and ranked thousands of Python code samples for **RLHF pipelines**; built quality rubrics for frontier AI models.
- Developed prompting strategies and evaluation criteria measuring correctness, efficiency, and edge case handling.

Automotive Cybersecurity Intern

Aug 2024 – Oct 2024

Swift-ACT

- Built security scenario catalogs with threat models and mitigations for **CAN bus** and **ECU** communication systems.
- Ran attack simulations (spoofing, replay, DoS) in **Dyna4** and **CARLA** against in-vehicle network architectures.
- Used **Vector CANoe** for CAN simulation, fault injection, and ECU validation; fuzz-tested **AEB** braking components.

Selected Projects

🔗 [Case studies and live demos at romia.dev](https://romia.dev)

Talos: Multi-User RAG Document Platform

Graduation Project (Team), Ongoing

- Owned the ingestion and retrieval core of a **FastAPI** platform for chatting with your own documents, with cited answers streamed over **SSE**.
- Built async ingestion (**ARQ** worker, **MinIO**, **Milvus**) with a race-safe processing state machine and **Google Drive** import.
- Two-stage retrieval: dense + **BM25** hybrid with reciprocal rank fusion, then cross-encoder reranking; built a statistical evaluation harness.

Virtual Banking Microservices

 [GitHub](#)

- Architected event-driven microservices with **Spring Boot** and **Kafka** for banking operations with OAuth2 security.
- Integrated a **LangChain**-powered AI agent as an independently scalable microservice for conversational banking.

ContextIQ: Production RAG System

</> [Demo](#)  [GitHub](#)

FastAPI RAG API supporting 11+ file formats with hybrid retrieval (TF-IDF + vector) and **ChromaDB** vector search.

Multimodal Emotion Recognition

</> [Demo](#)  [GitHub](#)

Cross-modal attention model fusing **HuBERT** audio and **EfficientNet** visual encoders with bidirectional fusion (8 classes).

Other: [VoicePrint](#) (Sentence-BERT style transfer) · [Connect4 AI](#) (minimax, alpha-beta) · [CSP Sudoku Solver](#) (AC-3, MRV)

Education

Alexandria University, Faculty of Engineering

Expected Jul 2026

B.Sc. in Computer & Communications Engineering (AI Concentration) · Cumulative GPA: 3.73 / 4.0

Technical Skills

Languages Python, Java, JavaScript/TypeScript, SQL, C/C++
Backend Spring Boot, FastAPI, Microservices, Kafka, REST APIs, OOP, Design Patterns, Webhooks, PostgreSQL, Redis
Frontend React, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Design
AI / ML LangGraph, LangChain, PyTorch, RAG, Vector DBs (ChromaDB, pgvector), Computer Vision, NLP
DevOps Docker, Git, GitHub Actions, CI/CD, Railway, Linux, Bash
Certifications: micro1 Certified Software Engineer · [Google AI Agents](#) · Meta Back-end · AWS Cloud & ML
Competitions: MTC-AIC3 BCI (Egypt National AI) · ECPC (3x) · Hepsiburada NLP Hackathon (TEKNOFEST)
Languages: English (Fluent) · Arabic (Native) · German (A2) · Spanish (Conversational)